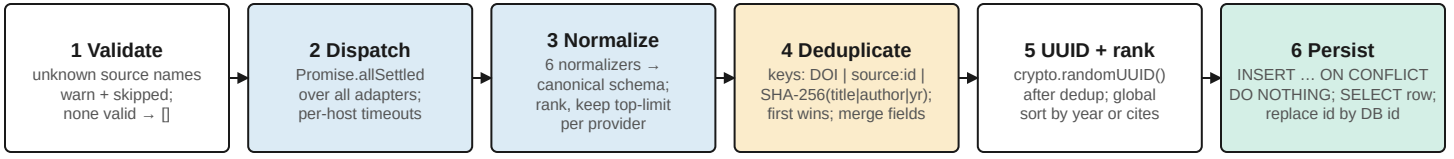

Stage 2 detail — one adapter per provider (all issued concurrently; the search returns when every adapter has settled)

	requests issued per search	timeout	
`openalex`	1 GET per_page = 3·limit (relevance)	15 s	
`arxiv`	1 GET 3·limit; adapter-level 3 s throttle	30 s	HTTP error / timeout → [ ] (reported as success)
`semantic_scholar`	1 GET limit; one retry after 2 s on 429	15 s	
`crossref`	1 GET rows = limit; type filter	15 s	
`europe_pmc`	1 GET + up to limit Crossref look-ups for PDF	15 s / 3 s	
`pubmed`	ESearch → 350 ms spacing → EFetch (XML)	25 s	

Downstream use of the result list

PDF acquisition (optional)
fetch(pdf_url) → %PDF- header check → temp file → copy to raw_papers/<title>--<uuid>.pdf → UPDATE papers.local_pdf_path

AI analysis (user-triggered, serial)
ai:analyzePaper per paper → Groq POST prompt = title + abstract + free-text task; answer written to analysis/<title>.md

Library view
scan raw_papers/ + SELECT * FROM papers; match by UUID suffix, stored path or source_id substring; else 'External File'